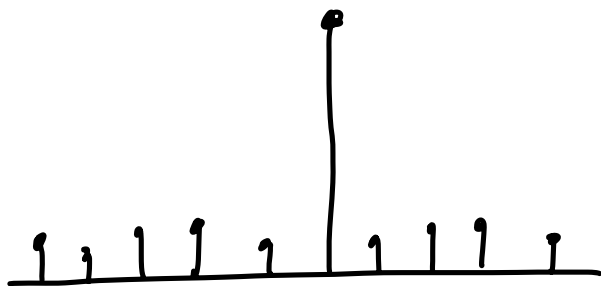


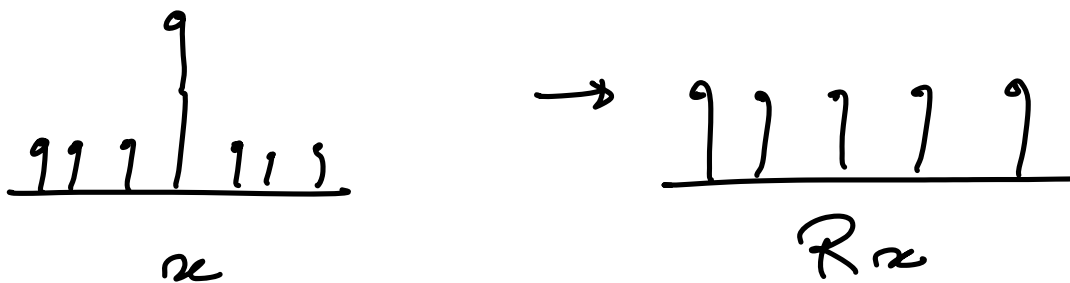
What if outliers are important?



E.g. Activation tensor. are being quantised.

One approach: Separately store a sparse mask.

Alternatively "Rotate the Vector" to distribute the energy



Quantisation: $\underbrace{\Phi(\vec{x})}_{\downarrow \text{Quantisation of vector}} \rightarrow \underbrace{\Phi(R\vec{x})}_{\downarrow \text{Element wise quantisation.}}$

* we can use linear quantisation for Φ .

How do we choose a rotation matrix.

* If $\vec{x}$ was fixed / unknown?

e.g: [fixed] weight matrix

e.g: [unknown] Activation matrix

Say, $\vec{x}$ was fixed.

A. Determine R which is optimal.

$\Rightarrow$ But now you have to store R or $R^{-1} \in \mathbb{R}^{d \times d}$

R : needs to be structured so that -

A given structured matrix may or may not work for a specific $\vec{x}$.

In fact for any chosen R , there is a $\vec{x}$ for which $R\vec{x}$ is catastrophic w.r.t quantisation!!

$\rightarrow$ Example?

What should we do?

[Idea] Randomisation

R : Random : So that with high probability, Rx is not adversarial.

Structured :- It can be stored efficiently.

Example of commonly used transforms

$\Rightarrow$ Randomised Hadamard.

$$H_{nn} = \begin{pmatrix} H_{n-1} & H_{n-1} \\ H_{n-1} & -H_{n-1} \end{pmatrix} \quad H_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$R = \underset{\substack{\downarrow \\ \text{Random Sign vector}}}{\text{Diag}(\vec{S})} \cdot H_{nn}$$

$S_i \sim \{\pm 1\}$ with equal probability.

Why hadamard : "Computational Comp"

Multiplication is $O(n^3) \rightarrow O(n^2 \log n)$

"Hada Core" : Kernel for Hadamard transform. [Add link]

Memory Cost

1 bit per element.

* Note: we can avoid this $O(n)$ memory by using Universal Hash functions. (More on that later)

Analysing effect of Rotations
on outliers

For simplicity Consider

$$R = \begin{bmatrix} \vdots \\ \dots N(0, \frac{1}{\sqrt{n}}) \dots \\ \vdots \end{bmatrix}$$

$$R_{ij} \underset{iid}{\sim} N(0, \frac{1}{\sqrt{n}})$$

Random Gaussian Matrix is
almost orthogonal

$$\|Rx\|_2 = (1 \pm \epsilon) \|x\|_2$$

with high probability

Given x , what is the
probability that some
co-ordinate of Rx is greater
than M .

$$P((Rx)_i > M) = ?$$

$$y = Rx$$

$$y_i = \sum_j r_{ij} \cdot x_j$$

$$E(y_i) = 0$$

$$\text{Var}(y_i) = \sum x_j^2 \cdot \text{Var}(r_{ij})$$

$$\text{Var}(y_i) = \|x\|_2^2 \cdot \frac{1}{n}$$

$$\text{Var}(y_i) \propto \frac{1}{n}$$

y_i = Sum of Gaussian
Variables

$$y_i \underset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \frac{\|x\|_2^2}{n}\right)$$

$$P_x(|y_i| > M) = 2 \underset{\hat{c}_{\text{CDF}}}{\Phi}^{-1}(M)$$